

ROBINPREET SINGH CHAHAL

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Education

Chitkara University

Bachelor of Engineering (B.E.) in Computer Science Engineering

2024 – 2028

CGPA: 8.64/10

Projects

PulseMQ | Java, Multithreading, Concurrent Queues

[GitHub](#)

- Built a Kafka-inspired event streaming engine using producer-consumer architecture and asynchronous message processing.
- Implemented topic-based messaging with concurrent event delivery across multiple consumers.
- Designed thread-safe queue management and event dispatching for scalable communication.

Route Optimization Engine | Java, Graph Algorithms

[GitHub](#)

- Developed an intelligent route optimization system using Dijkstra's shortest path algorithm.
- Modeled dynamic road networks as weighted graphs for efficient pathfinding.
- Implemented route computation and network traversal for transportation scenarios.

RateLimitEngine | Java, Backend Systems

[GitHub](#)

- Built an API rate limiter using the Sliding Window algorithm to control request throughput.
- Simulated backend throttling systems used by large-scale web services.
- Implemented request tracking and window-based enforcement mechanisms.

JSync Subscription Engine | Java, OOP, System Design

[GitHub](#)

- Developed a SaaS subscription and billing engine inspired by platforms such as Netflix and Spotify.
- Implemented auto-renewals, billing cycles, discount handling, and payment workflows.
- Designed extensible object-oriented architecture for subscription management.

AgriSense AI | Python, Flask, Machine Learning

[GitHub](#)

- Built a crop recommendation platform using soil parameters and weather conditions.
- Integrated machine learning models for crop recommendation based on soil nutrients, rainfall, and environmental conditions.
- Used historical mandi price data to support data-driven agricultural decisions.

Technical Skills

Programming Languages: Java, MySQL, Python, C++

Core Java: OOP, Collections Framework, Exception Handling, Multithreading, JDBC

Database: MySQL

Computer Science Fundamentals: Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks

Tools: Git, GitHub, VS Code, Flask

Certifications

- Oracle Cloud Infrastructure 2025 AI Foundations Associate (1Z0-1122-25)
- Microsoft Certified: Azure AI Fundamentals (AI-900)
- Microsoft Certified: Azure Fundamentals (AZ-900)
- Microsoft Certified: Azure Data Fundamentals (DP-900)
- Statistics Foundations – Meta
- Harness Git and GitHub – Open Source Chandigarh